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Predictive genetic circuit design for phenotype reprogramming in plants

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Plants, with intricate molecular networks for environmental adaptation, offer groundbreaking potential for reprogramming with predictive genetic circuits. However, realizing this goal is challenging due to the long cultivation cycle of plants, as well as the lack of reproducible, quantitative methods and well-characterized genetic parts. Here, we establish a rapid (~10 days), quantitative, and predictive framework in plants. A group of orthogonal sensors, modular synthetic promoters, and NOT gates are constructed and quantitatively characterized. A predictive model is developed to predict the designed circuits' behavior accurately. Our versatile and robust framework, validated by constructing 21 two-input circuits with high prediction accuracy ($R^2 = 0.81$), enables multi-state phenotype control in both *Arabidopsis thaliana* and *Nicotiana benthamiana* in response to chemical inducers. Our study achieves predictable design and application of synthetic circuits in plants, offering valuable tools for the rapid engineering of plant traits in biotechnology and agriculture.

Plants, unlike animals, are usually immobile and must cope with unfavorable conditions. They may have developed the most unique and sophisticated molecular networks that allow them to sense, respond, and adapt to their surrounding environments. Rewiring these networks in predictive and quantitative ways for precise phenotype manipulation and reprogramming in plants is, therefore, a highly desirable goal[1], which will not only advance our understanding of plant biology[2,3], but also address the increasing challenges in plant biotechnology and green agriculture[4,5].

Synthetic molecular networks or genetic circuits can process and manipulate information in living organisms, creating novel and useful biological functions in response to environmental signals[6–8]. Although predictive gene circuit design has been implemented in microorganisms such as bacteria and yeast, it remains a challenge in more complex organisms such as plants[9–12]. Plants have a long cultivation cycle and lack reliable and quantitative methods for characterizing genetic parts, leading to a scarcity of well-defined components and a perception that plants defy rational engineering[12,13]. Existing designs often require multiple iterations of the Design-Build-Test-Learn cycle to achieve optimal performance, which demands specialized expertise and significant time investments (>2 months for each iteration)[14,15]. The few logic circuit designs that were built in plants often lack scalability and quantitative modeling in the design process, making the procedure inefficient[16–20]. Moreover, current genetic circuits in plants are often hard-coded for each state, and switching between states requires transforming different DNA sequences[16,19]. These limitations significantly impede the development of robust and predictive genetic circuit design in plants for phenotype manipulation and reprogramming.

In this work, we develop a rational and predictive genetic circuit design framework in plants (Fig. 1a). We establish a robust transient expression system and adapt the concept of relative promoter unit (RPU), enabling rapid (~10 days), reproducible, and quantitative characterization of genetic parts and synthetic circuits (Fig. 1b). We construct and quantitatively characterize a library of orthogonal sensors and NOT gates. A predictive modeling approach is then developed to simulate the behavior of the designed circuits. We demonstrate the versatility and robustness of our framework by constructing 21 two-

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input genetic circuits with various logic functions (14 types) and show that all the predicted performances of the circuits are in close agreement with the experimental data ($R^2 = 0.81$). These designed circuits can be used to program multi-state phenotypes in response to input chemical signals for multi-state phenotype control in both *Arabidopsis thaliana* and *Nicotiana benthamiana* (Fig. 1c).

Results

Establishing a rapid and robust quantification method using RPU

Predictive genetic circuit design in plants requires a rapid and reliable method to accurately quantify the performance of genetic parts. Transient expression methods, such as protoplast transfection, introduce genes into cells for a limited duration, allowing rapid testing of genetic parts without the need for stable genetic transformation. However, high variability between batches in protoplast transient expression has been reported[13], and precise quantification of genetic parts in such transient systems has not been established to facilitate gate normalization and connection for circuit design. We first set up the *Arabidopsis* leaf mesophyll protoplast transfection system to quantify several native promoters with different strengths, from low (NOS, ACT2) to high (UBQ10, 35S), using firefly luciferase (LUC) as the reporter (Supplementary Fig. 1a). We also constructed and tested three truncated promoters by truncating the strongest 35S promoter, including 200 bp, 100 bp, and the core promoter with only the TATA box (Supplementary Fig. 1b). The LUC outputs of native promoters showed the expected trends consistent with previous reports[21]. For the truncated promoters, we found that the 200-bp promoter retained its high strength, while the other two promoters showed a reduction in activity (Supplementary Fig. 1b). However, the variations in LUC outputs were quite obvious (Supplementary Fig. 1a, b).

To reduce this variation, we incorporated a normalization module featuring a β-glucuronidase (GUS) protein driven by the 200-bp 35S promoter. In this pipeline, each plasmid contained two modules: a GUS normalization module and a LUC circuit module. The LUC/GUS ratio provided normalized values for the output of each construct. After normalization, we not only observed a consistent trend in all seven promoters (Supplementary Fig. 1c, d) but also a reduction in variations (Supplementary Fig. 1e). However, we still observed large variation in the LUC/GUS values when comparing the same promoter across different batches of protoplasts and in different experimental setups (Fig. 1d), consistent with the batch variation reported previously[13].

To overcome the issue, we used the design principle of standardization and adapted the concept of relative promoter units (RPUs) into our system[22]. RPUs provide a relative measure of promoter strength compared to a reference promoter within each protoplast batch. We chose the 200-bp 35S promoter as the reference promoter, defining its LUC/GUS value in each batch as 1 RPU. After converting the measurements to RPUs, the batch and experimental setup variation was significantly reduced, enabling more reproducible and comparative analyses (Fig. 1d, e). In all subsequent experiments, we used this established pipeline and RPUs to quantify the output of our designed genetic parts and circuits.

Designing modular synthetic promoters with repressible activity

To expand the genetic part library in plants, we aimed to design modular and synthetic promoters (P_{syn}) that can be repressed by specific repressors, which can then be used to build NOT gates for genetic circuit design. We chose the strong constitutive 200-bp 35S promoter as the backbone for our P_{syn} (Fig. 2a). This promoter has the 35S core promoter and the 35S core enhancer, with specific subdomains (A1 and B1-3) contributing to promoter activity[23]. The TATA box in the core promoter and the CAAT box (CCAAT/CAT box) in subdomain A1 are conserved motifs in the 35S promoter and other eukaryotic promoters. Downstream of the transcription start site (TSS), we added the 5′-leader sequence (Ω) of tobacco mosaic virus (TMV) before the start codon (ATG) to enhance translation of the coding gene[24]. We then replaced the DNA sequence with either one or two copies of operators of the TetR family repressor at five different sites of the 200-bp 35S promoter, maintaining the overall length of the promoter sequence (Fig. 2a and Supplementary Table 1). These repressors can recognize and bind to specific operator sequences within a promoter region and block RNA polymerase from transcribing downstream gene expression. To evaluate the repression ability of different promoters, we added a nuclear localization signal (NLS) at the C-terminal of the repressor and co-expressed both the promoter and the repressor on the same plasmid (Fig. 2b).

We first used the LmrA repressor and its cognate operator to construct and test synthetic promoters (Fig. 2c). In the case of one-copy designs, replacing the operator at position 5 (LmrRe5), downstream of the TATA box, achieved the highest fold repression. Replacing the operator at position 3 (LmrRe3) significantly compromised the promoter strength, possibly due to the disruption of three adjacent CAAT boxes in subdomain A1, affecting the recognition of the conserved CAAT-box by native transcription factors[25]. We therefore excluded the replacement design at position 3 for subsequent designs. In the case of two-copies of operator replacement design, we observed the highest fold-repression for LmrRe45.

Next, we collected more repressors from the TetR family (PhlF, IcaR, BM3R1, SrpR, and BetI) that showed robust performance in other organisms such as bacteria[11] and yeast[10]. We constructed a library of synthetic promoter-repressor pairs to investigate their performance (Supplementary Fig. 2 and 3). The strength of all the P_{syn} promoters without repression was almost the same as that of the starting promoter. For these repressors, the promoters that generated the highest dynamic range were identified (Fig. 2d). In general, we found that placing the operators between the CAAT boxes and TSS achieved improved fold-repression (Fig. 2d, Supplementary Fig. 2 and 3). In these designed promoters, the fold repression ranged from 4.3 (IcaR) to 847 (PhlF). Furthermore, we evaluated the orthogonality of the repressor-P_{syn} set and observed that they exhibited high levels of orthogonality with minimal crosstalk between BM3R1 and PhlF (Fig. 2e). Interestingly, the off-target interactions we observed showed some differences compared to those in *E. coli*[11], mammalian cells[26], or *Nicotiana benthamiana*[16]. This discrepancy may be attributed to differences in the expression levels of transcription factors, the presence of multiple operators in synthetic promoters, and the ability of VP16/repressors to recruit or inhibit the transcriptional machinery even when delivered to a promoter with low affinity.

Quantifying the input–output parameters of sensors and NOT gates

Accurate quantification of input-output characteristics for genetic parts is vital for constructing circuits with predictable functions. Sensors can provide continuous titration to measure and parameterize the input-output response of NOT gates for predictive genetic circuit design. We first measured the input-output characteristics of two sensors, an auxin sensor and a cytokinin sensor, using the transient expression system in protoplasts that we established.

The native promoter of the auxin-inducible GH3.3 gene (At2g23170) was chosen as the auxin sensor[27,28]. Our results showed that the 300 bp promoter of GH3.3 is highly sensitive to auxin (1-naphthaleneacetic acid, NAA) levels ranging from 0 to 1.2 μM (Fig. 3a). We fitted the response function of the sensor to a Hill equation. The fold induction of the auxin sensor was 40 with a Hill coefficient n of 1.32 (Supplementary Table 2). The two-component-output sensor new (TCSn), which harbors the concatemerized B-type *Arabidopsis* response regulator (ARR)-binding motifs and a minimal 35S promoter[27,29], was chosen as the cytokinin sensor. The dynamic range

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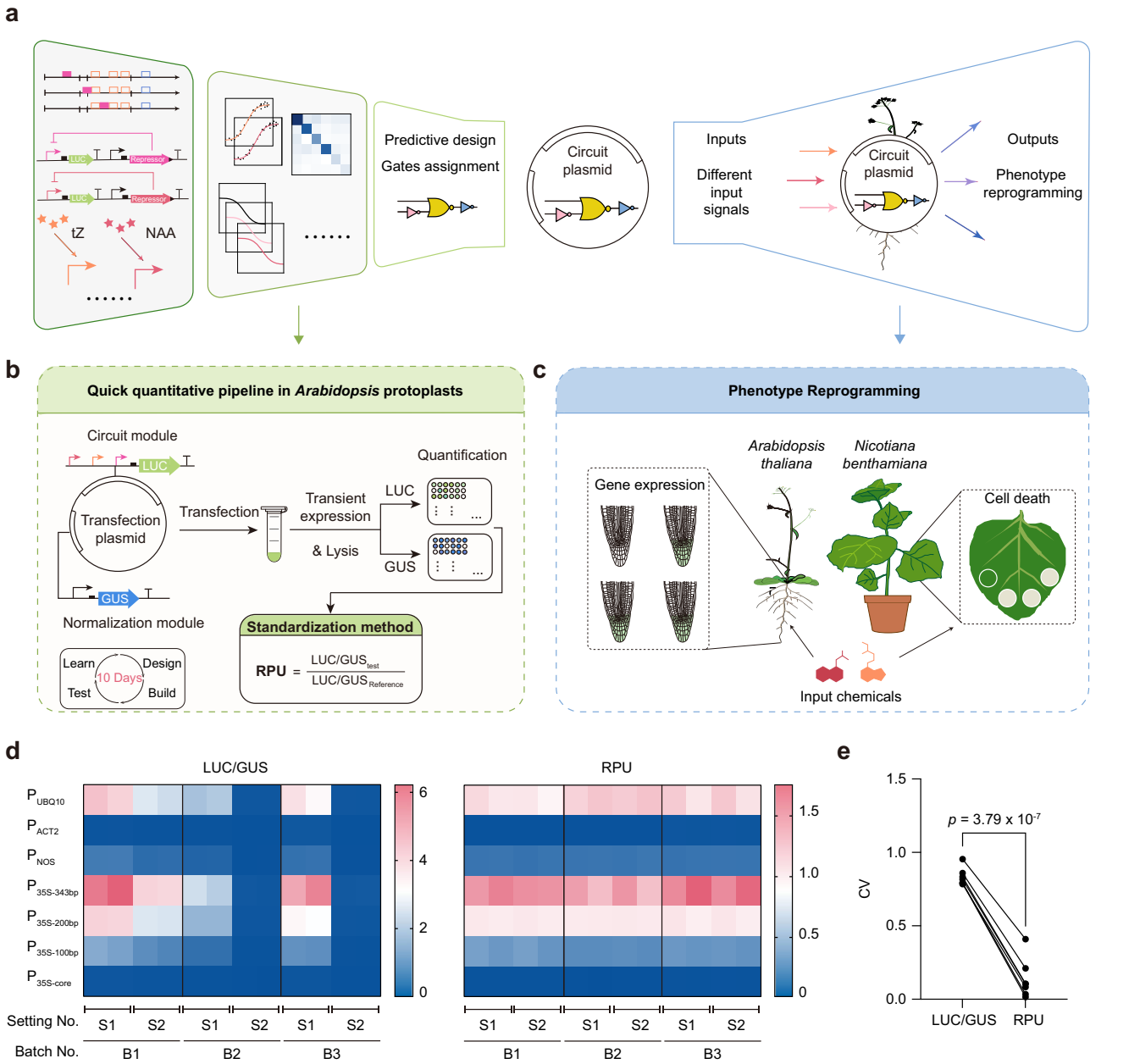


Fig. 1 | Overview of the predictive genetic circuit design and applications framework. a Framework of predictive genetic circuits design and downstream applications. **b** A rapid and reproducible method for quantifying genetic circuits in *Arabidopsis* protoplasts. **c** Schematic of the application of genetic circuits for phenotype reprogramming. **d**, Constitutive promoter activities in different protoplast batches and equipment settings, measured by LUC/GUS ratios and relative promoter units (RPUs). Three biological replicates (different protoplast batches, B1, B2, B3) of protoplasts were tested under two LUC gain settings (S1, S2), with two technical replicates per setting for each biological replicate. **e** Coefficients of variation (CVs) for LUC/GUS and RPU measurements of the seven promoters. $p = 3.79 \times 10^{-7}$ by Student's *t* tests, two-tailed, and paired. *Arabidopsis* flowering plant illustration is adapted from an image by Frédéric Bouché on Figshare (https://doi.org/10.6084/m9.figshare.7159937), and licensed under the Creative Commons Attribution 4.0 International (CC BY 4.0, https://creativecommons.org/licenses/by/4.0/). Outlines were partially modified in the adaptation. Source data are provided as a Source Data file.

of the cytokinin sensor was smaller compared to the auxin sensor, with a higher OFF state (0 nM) and a lower ON state (300 nM), resulting in a fold induction of 12 and a Hill coefficient *n* of 0.8 (Fig. 3b and Supplementary Table 2). In addition, we found that both sensors selectively responded to their own specific molecules, demonstrating good orthogonality for genetic circuit design (Fig. 3c).

Next, candidates from the orthogonal repressor-P_{syn} set were coupled with the auxin sensor $P_{GH3.3}$ to measure the parameters of NOT gates (Fig. 3d). We transfected the sensor plasmids and NOT plasmids in the same batch with identical NAA induction concentrations and both were measured in RPUs, such that the output of the sensor can directly serve as the input for the NOT gates. The transfer function of each NOT gate was determined by fitting the input-output relationship to the Hill function (Fig. 3e). The varied performance of NOT gates within the LmrA, BM3R1, PhlF, SrpR, BetI and IcaR groups contributes to the diversity of our NOT gates library for various applications, where different gene expression levels are desired (Fig. 3e and Supplementary Fig. 4). Hill coefficients are crucial for robust gate connection and circuit design. In our study, we found that most of the NOT gates exhibited a Hill coefficient *n* greater than 2 (Supplementary Table 3). A higher Hill coefficient means the system exhibits a steep, nonlinear response to input signals. And a Hill

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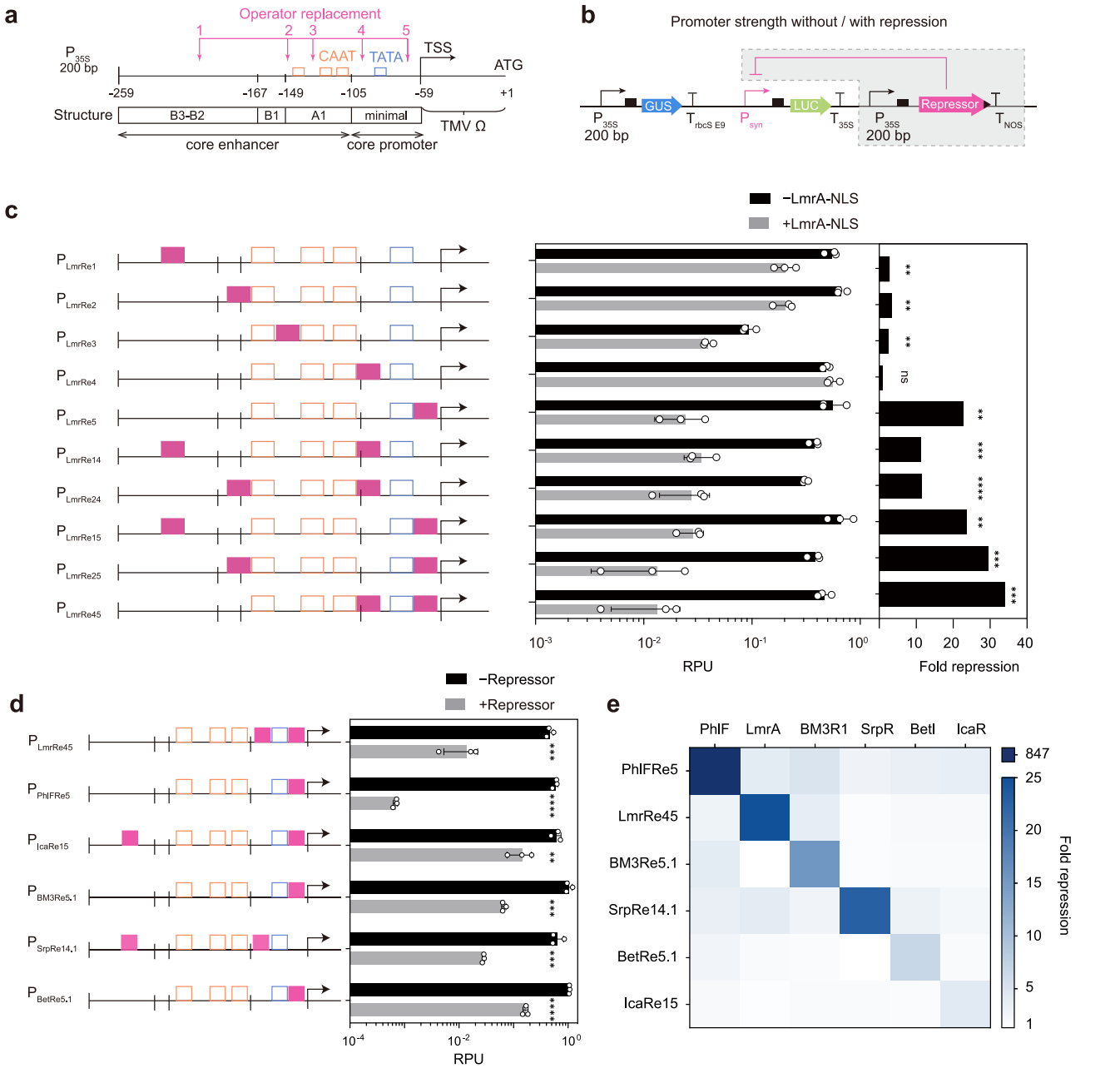


Fig. 2 | Design of operator-based synthetic promoters in *Arabidopsis*. **a** Schematic of the synthetic promoter designs based on the 200-bp 35S promoter. Five sites for operator replacement are highlighted in magenta. TSS, transcription start site. **b** Schematic of plasmids for screening synthetic promoters. Tobacco mosaic virus (TMV) *Ω* sequence is represented by the little black square, nuclear localization signal (NLS) is represented by the black triangle after the repressor. **c** Design and screening of synthetic promoters based on the LmrA operator. **d** A library of synthetic promoters and their strength after repression. In (**c**) and (**d**), black bars indicate RPUs of synthetic promoters, the RPUs after repression are shown in gray bars. Data are mean ± standard deviation of three independent experiments, $^{****}p < 10^{-4}$, $^{***}p < 10^{-3}$, $^{**}p < 10^{-2}$, ns (not significant) by Student's *t* tests, two-tailed, and unpaired. **e** Orthogonality of synthetic promoters. Data are mean values of three biological replicates. Source data are provided as a Source Data file.

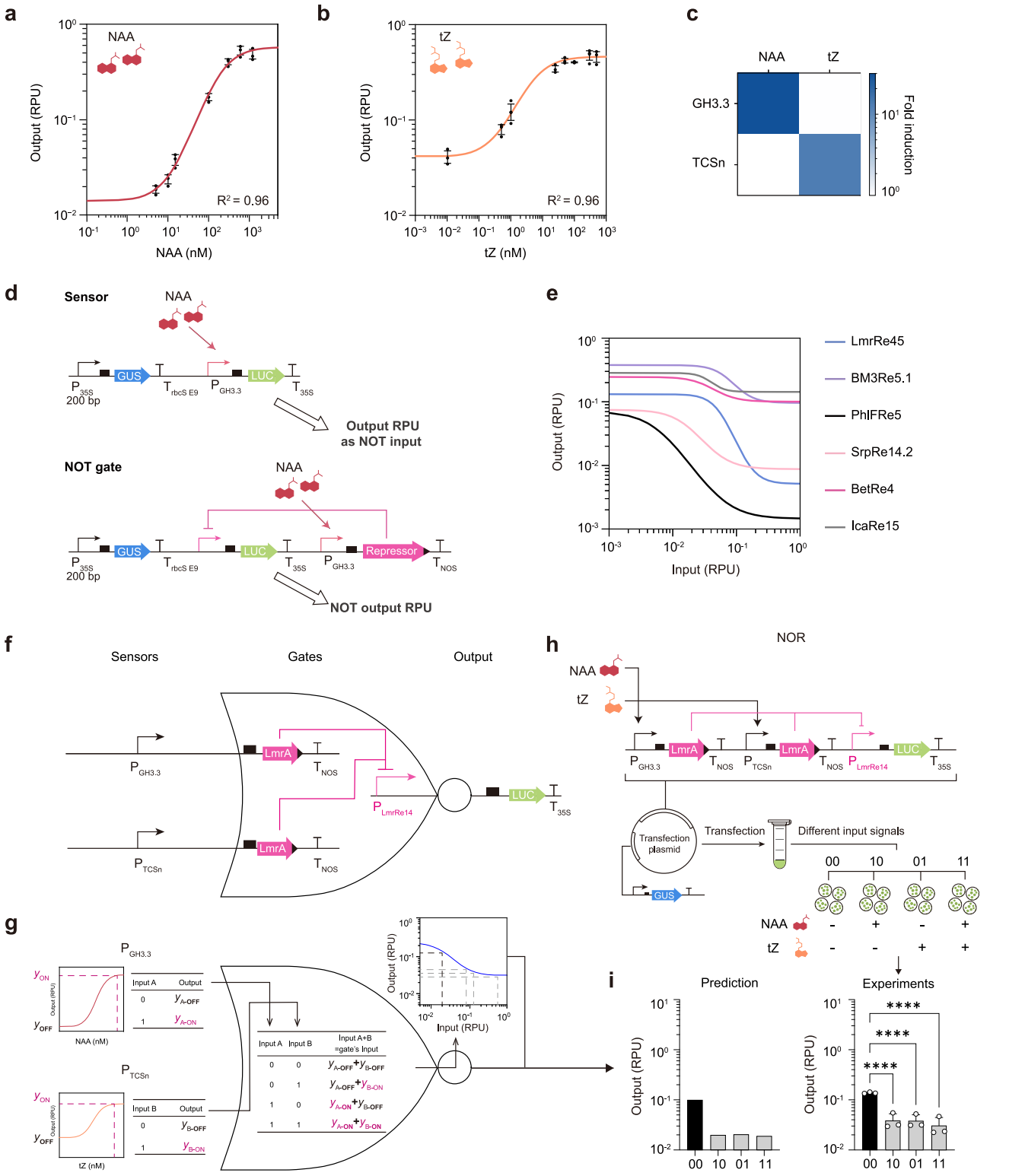
coefficient greater than 2 ensures effective signal amplification of the input signals, which is critical for maintaining reliable signal propagation in multi-layered circuit designs[30].

Predicting NOR gate performance using a quantitative modeling approach

Our parameterized sensor and NOT gate library enable the construction of various Boolean genetic circuits with different logic functions that can respond to different input signals. We next developed a predictive modeling approach that utilizes the parameters of different parts to accurately predict genetic circuit performance. The modeling was carried out following the topological structure of the circuit. In the case of "series" connections, the output RPU of the previous layer was used as the input for the next layer. For "parallel" connections, RPU values were simply added together to calculate the combined input, without introducing any additional fitting parameters.

The programmability of the NOR gate in response to both endogenous and exogenous signals is crucial for achieving sophisticated logic control of gene expression in plants. Therefore, we tested our predictive modeling with the modular design and prediction of NOR gates (Fig. 3f, g). In the modular design (Fig. 3f), the auxin and cytokinin sensors served as the inputs of the NOR gate. Each of the sensor

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promoters drove the expression of an individual copy of the repressor. The P_{syn}, in this case, $P_{LmrRe14}$, was repressed when either or both copies of the repressor were expressed, and the activity of P_{syn} served as the output of the NOR gate. In the predictive modeling (Fig. 3g), the two sensors are in "parallel", so the RPU values (both ON and OFF states) of each sensor are summed to reflect the yield of cellular repressors. Specifically, we considered the following input combinations: OFF (A, $P_{GH3.3}$) + OFF (B, P_{TCSn}), ON (A) + OFF (B), OFF (A) + ON (B), and ON (A) + ON (B), which correspond to the binary inputs 00, 10, 01, and 11, respectively. The repressors and $P_{LmrRe14}$::LUC output are in "series" connections, so the yield of cellular repressors feeds as the input of the LmrRe14 NOT gate. This input is then mapped onto the Hill function of the LmrRe14 NOT gate to calculate the output (y values of the NOT gate), which is also the final output of the NOR gate. During the design process, different NOT gates can be selected to construct different NOR gates (Fig. 3e, g). Using this approach, we predicted and constructed a NOR gate. We observed that the expression patterns in prediction and experiments were closely matched (Fig. 3h, i).

Fig. 3 | Quantitative characterization of sensors and NOT gates, and predictive design of NOR gates. Response functions of naphthaleneacetic acid (NAA, **a**) and trans-zeatin (tZ, **b**) sensors. NAA concentrations were 0 (not shown), 5, 10, 15, 100, 300, 600, 1200 nM, and tZ concentrations were 0 (not shown), 0.01, 0.5, 1, 25, 50, 100, 300, 500 nM. The response function was fitted using the Hill equation, $R^2 = 0.96$ for both NAA (**a**) and tZ sensor (**b**). Data are mean ± standard deviation of three independent experiments. **c** Orthogonality of NAA and tZ sensors. Fold induction was calculated as RPU (+inducer)/RPU (−inducer) for 1200 nM NAA and 300 nM tZ. Results are the mean values of three biological replicates. **d** Schematic of parameterizing input and output RPUs for NOT gates. Input promoter activities were measured using the NAA sensor plasmids in the same batch of protoplasts. **e** Response curves of six NOT gates fitted using the Hill equation. More details are provided in Supplementary Fig. 4. **f** Modular design of the LmrRe14 NOR gate. **g** Modeling of NOR gate. Two sensors' inputs are in "parallel", RPUs (both ON and OFF states) of each sensor are summed to reflect the yield of cellular repressors: OFF (A) + OFF (B), ON (A) + OFF (B), OFF (A) + ON (B), and ON (A) + ON (B). The repressors and $P_{LmrRe14}$::LUC output are in "series" connections, RPUs of cellular repressors feed as the input of the LmrRe14 NOT gate. This input is then mapped onto the Hill function of the LmrRe14 NOT gate to calculate the output (y values of NOT gate), which is also the final output of the NOR gate. Input A: NAA, Input B: tz. **h** Schematic of testing the same circuit under different input signals. **i** NOR gate performance in prediction and experiments. Data are mean ± standard deviation of three independent experiments, $^{****}p < 10^{-4}$ by One-way ANOVA with Tukey's multiple comparisons test. Source data are provided as a Source Data file.

Predictive design of complex genetic circuits

To further evaluate the performance of genetic circuits constructed using our identified genetic parts and the predictive ability of the modeling approach, we designed, predicted, constructed, and measured the performance of 20 additional 2-input genetic circuits with various logic operations (Fig. 4a and Supplementary Fig. 5a–c). For all these circuits we designed, we observed that the predicted and measured outputs for different states within and across different circuits were in good agreement. There were some minor differences in RPUs for the more complex circuits with multiple layers, such as NIMPLY and AND (Fig. 4a). These discrepancies could be attributed to three main factors[31,32]: (1) cellular resource competition, as complex circuits demand substantial transcriptional and translational resources, potentially leading to signal attenuation[33]; (2) protein turnover dynamics, where non-dividing protoplast system permits the accumulation of repressors, hindering effective signal propagation in layered circuits[34]; and (3) simplifications in our predictive modeling, such as ignoring resource constraints and degradation dynamics, which may have led to inaccuracies, particularly for more complex, multi-layer circuits[35].

To evaluate the predictive ability of circuits' performance, a parameter called circuit score (S) was used as the performance evaluation metric, which is defined as the ratio of the lowest predicted ON state to the highest predicted OFF state of the circuit output[8]. This metric allows us to compare and assess the performance of different circuits. We observed the predicted S and experimentally measured S showed a high correlation (Pearson $r = 0.90$, $R^2 = 0.81$) among circuits incorporating two sensors as inputs (Fig. 4b). The overall high agreement between the predicted and experimental outputs demonstrated the robustness and effectiveness of our approach to evaluate the designed circuits in plants.

Multi-state gene expression logic control in vivo

With the successful predictive design of genetic circuits in protoplasts, we next aimed to verify whether the genetic circuits we constructed could program gene expression in vivo in response to input chemical signals as expected. We replaced the LUC output of the circuits with GFP for visual observation in vivo and implemented the circuit in stable transgenic *Arabidopsis* living plants to observe the gene expression pattern in plant roots (Fig. 5a).

We assessed the performance of A IMPLY B and NAND circuits in vivo. For both circuits, we observed the expected and input-responsive logical behavior within *Arabidopsis* roots of three T2 lines for A IMPLY B and four for NAND (Fig. 5b–g and Supplementary Fig. 6a–e). We next quantify the relative expression levels of GFP in vivo. We focused on the root tip region, including mainly columella, quiescent center (QC) and lateral root cap (Supplementary Fig. 7c), where auxin and cytokinin responses are strong[29,36]. Remarkably, for both A IMPLY B and NAND circuits, we observed good consistency between the measured output results in protoplasts and the root data (Fig. 5d, g), highlighting the robustness and potency of the design framework developed in our study for in vivo applications.

Programmable control of plant hypersensitive response and cell death in tobacco

We next assessed the performance of our genetic parts and circuits using a nucleotide-binding leucine-rich repeat receptor (NLR) gene as an output to control plant hypersensitive response (HR) in *N. benthamiana*. In response to pathogen attacks, plants have developed a defense system utilizing intracellular receptors known as NLRs. By triggering immunity against pathogens, NLRs initiate the HR and induce host cell death, which serves to restrict pathogen growth and limit infection[37].

To control cell death, we conducted agroinfiltration experiments in *N. benthamiana* to measure the hypersensitive cell death response with different input signals. *Agrobacterium tumefaciens* containing the same circuit plasmids were infiltrated into four circular areas on the same tobacco leaf. 24 h after agroinfiltration, we induced them with different input chemical molecules and the phenotypic cell death was recorded after another 24 h (Fig. 6a). After induction, we also measured the amounts of electrolytes leaked from dying plant cells to quantify plant cell death[38,39] at different time points. Here, we selected the *Arabidopsis NRG1.1* (N REQUIREMENT GENE 1.1) autoactive allele *NRG1.1*^{D485V} as the activated NLR gene[40]. In the absence of pathogen attacks, autoactive NRG1.1^{D485V} oligomerizes and acts as a calcium-permeable channel that facilitates calcium influx and subsequently triggers cell death (Fig. 6b)[40–43]. In the TRUE gate, *NRG1.1*^{D485V} was driven by the 200-bp 35S promoter that is always ON regardless of the different inputs (Figs. 4a, 6c and Supplementary Fig. 7a). In contrast to the empty control, we observed fast plant cell death 24 h after agroinfiltration in the TRUE gate (Fig. 6d). The cell death also increased over time, demonstrating the effectiveness of NRG1.1^{D485V} in controlling plant hypersensitive response and cell death.

Next, we tested the OR circuit to control cell death using NRG1.1^{D485V} as the output. Our results showed that all states except "00" exhibited a strong immune response, which was observed under both white light and long-wave ultraviolet (Fig. 6e and Supplementary Fig. 7b). We observed increased cell death at 24 and 36 h after input induction (in 10, 01, and 11 states) compared to the 00 state, while no differences were observed at 12 h after induction (Fig. 6f). This suggests that the accumulation of the induced protein and the resulting cell death require at least 24 h to become evident. Furthermore, the quantified relative cell death in these 4 states closely matched our model prediction (Fig. 6g), showing a high correlation with Pearson $r = 0.92$ (Fig. 6h). This logical control of cell death in *N. benthamiana* demonstrates the potential for cross-species utility of our genetic parts and quantification results, as well as the ability to control cell death using chemical inducers and limit pathogen growth in future applications.

Discussion

Predictive design has always been a significant goal in the field of genetic circuit design, which depends on the quantitative characterization of genetic parts and rational modeling methods. The findings of our study suggest that, despite challenges associated with rational

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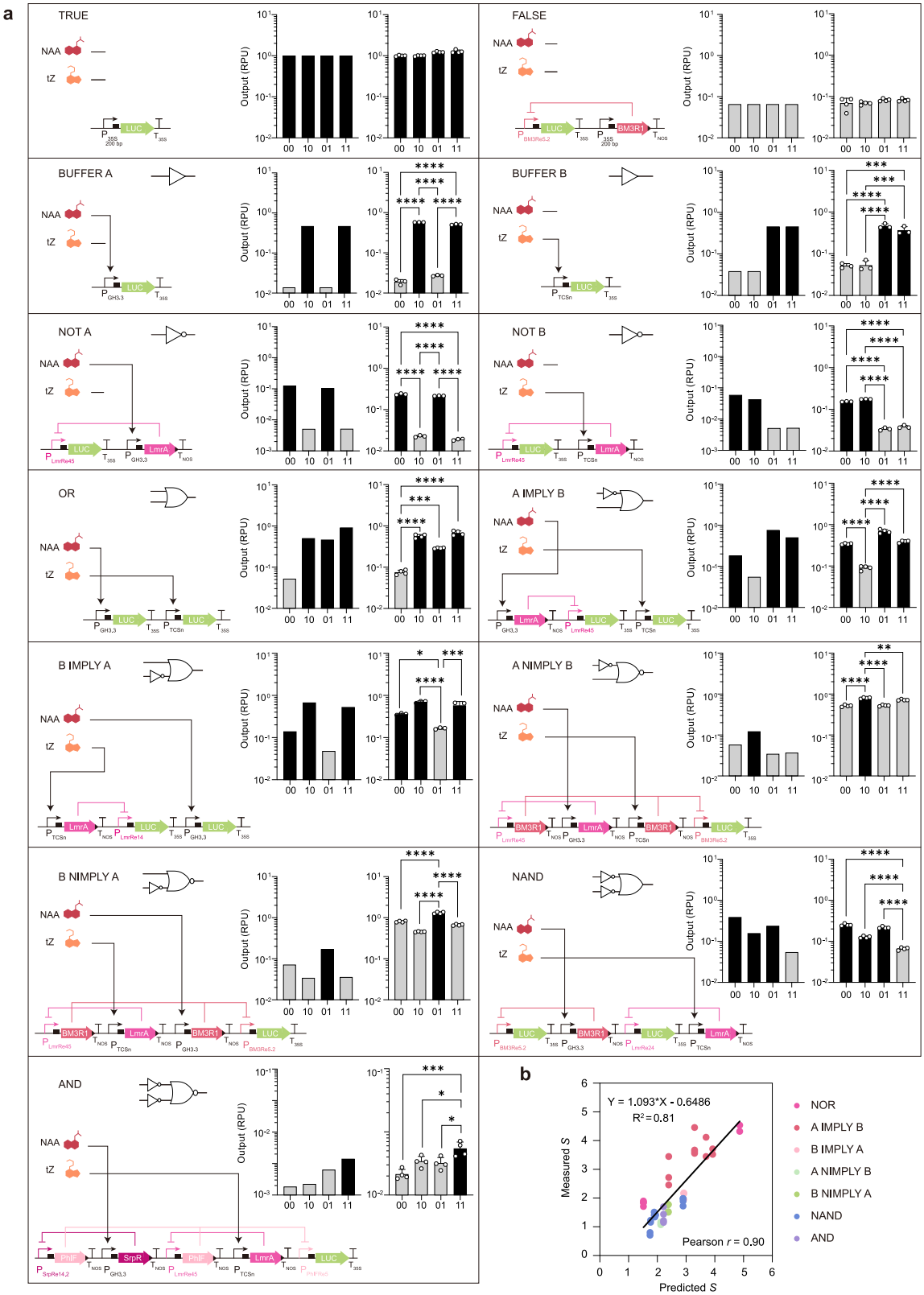


Fig. 4 | Predictive modeling and design of 2-input genetic circuits. a The design, prediction and actual experimental performance of 13 2-input genetic circuits. For each circuit, the left and right panels show the predicted and actual performance, respectively. The solid black and gray bars indicate the expected "ON" and "OFF" states of the circuit, respectively. Data are mean ± standard deviation of four (TRUE, FALSE, OR, A IMPLY B, NIMPLY, NAND and AND) or three (BUFFER, NOT and B IMPLY A) independent experiments. One-way ANOVA with Tukey's multiple comparisons were performed, with significance levels indicated as $^{****}p < 10^{-4}$, $^{***}p < 10^{-3}$, $^{**}p < 10^{-2}$, $^{*}p < 0.05$, ns (not significant). **b** Correlation between the predicted and actual performance of genetic circuit score (*S*) in the circuits incorporating two sensors as inputs. Each data point in the plot represents the predicted and actual *S* of a circuit, with different colors denoting different circuits. Source data are provided as a Source Data file.

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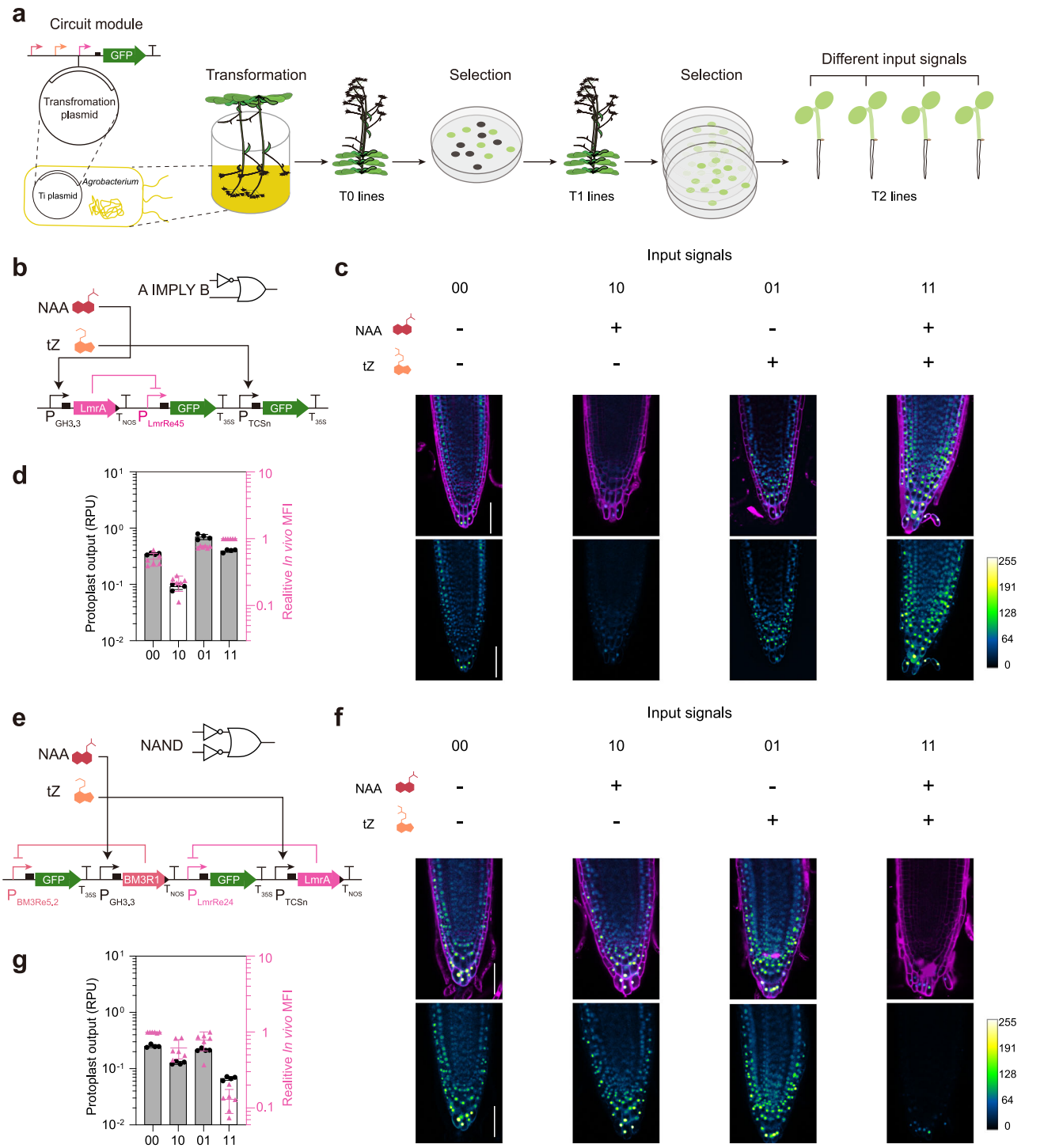
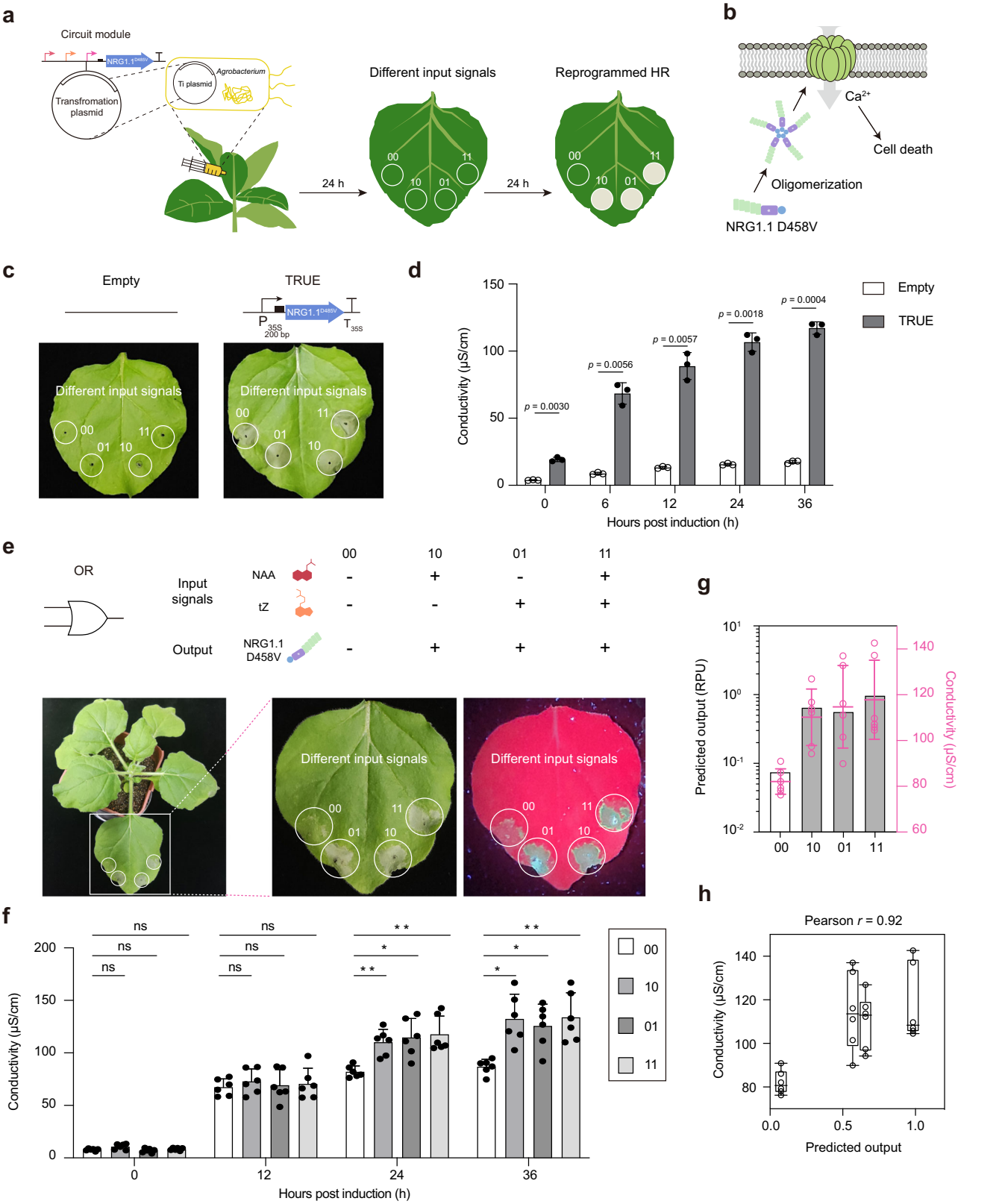


Fig. 5 | Input-responsive gene expression logic control in vivo by genetic circuits. a Schematic of multi-state gene expression control in stable transgenic lines in response to diverse input signals. Individual transgenic T2 lines were treated with different small molecules overnight and gene expression patterns in the root were observed by microscopy. Circuit designs of the A IMPLY B circuit (**b**) and the NAND circuit (**e**). The reporter gene is NLS-GFP. Confocal images of gene expression patterns of the A IMPLY B circuit (**c**) and the NAND circuit (**f**) in *Arabidopsis* root tips. Propidium iodide staining for cell boundaries is shown in magenta, GFP is mapped to the green-fire-blue color scale, and the scale bar is 50 µm. The input states are labeled at the top of each image, where the first digit indicates the presence (1) or absence (0) of NAA (1 µM) induction, and the second digit indicates the presence (1) or absence (0) of tZ (1 µM) induction. Relative GFP intensity in vivo of different states, overlaid with outputs of the A IMPLY B circuit (**d**) and the NAND circuit (**g**) measured in protoplasts (bars with black dots represent the mean ± standard deviation of four independent experiments). Pink triangles with error bars indicate mean relative GFP intensity with standard deviation, derived from six replicates for each gate across three T2 lines for A IMPLY B and four lines for NAND. Quantification details are provided in Supplementary Fig. 6 and Methods. *Arabidopsis* flowering plant illustration is adapted from an image by Frédéric Bouché on Figshare (https://doi.org/10.6084/m9.figshare.7159937), and licensed under the Creative Commons Attribution 4.0 International (CC BY 4.0, https://creativecommons.org/licenses/by/4.0/). Outlines were partially modified in the adaptation. Source data are provided as a Source Data file.

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engineering in plants[44–47], predictive design and modeling of genetic circuits are feasible when leveraging repressor-synthetic promoter pairs and NOT gates. Although some designs, such as the BUFFER and OR gates presented here, may appear relatively simple, they serve as foundational building blocks for more complex circuit construction. These basic logic gates provide essential proof-of-concept demonstrations, validating the application of logic gate principles in plant systems. These results mark an incremental yet significant step toward developing synthetic biology tools tailored for plant systems. The predictive modeling approach employed in our study proved to be highly effective in estimating the behavior of the designed circuits.

Encouragingly, we have also validated that the multi-state circuit performance in the stable transgenic plants matched with the circuit prediction with good agreement (Fig. 5d, g). This differs from the hard-

Fig. 6 | Programmable control of plant cell death. a Schematic of tobacco cell death regulated by circuits. The same genetic circuit was introduced through *Agrobacterium* infiltration, followed by inducers infiltration 24 h later. Conductivity was measured 24 h after inducer treatment. The input states are labeled at the top of each plot, where the first digit indicates the presence or absence of NAA (1 μM) induction, and the second digit indicates the presence or absence of tZ (1 μM) induction. **b** Schematic of cell death through the autoactivation allele *NRG1.1*^{D458V} oligomerization and Ca^{2+} influx. **c** Cell death triggered by the TRUE gate (NRG1.1^{D485V} overexpression) under white light or long-wave ultraviolet. Empty refers to *Agrobacterium* infiltration with an empty vector. **d** Quantification of cell death at 11 states in (**c**) by electrolyte leakage assay at different time points. Data are mean ± standard deviation of three independent experiments, statistical analysis of the results was performed by Student's *t* tests, two-tailed and unpaired. **e** Cell death controlled by the OR gate under white light or long-wave ultraviolet. **f** Quantification of cell death at four states in (**e**) by electrolyte leakage assay at different time points. Data are mean ± standard deviation of six biological replicates, $^{**}p < 10^{-2}$, $^{*}p < 0.05$, ns (not significant) by two-way ANOVA and Dunnett's multiple comparisons test. **g** Cell death quantification overlaid with model predictions in different states of the OR circuit (bars). Quantification of cell death at 24 h post-induction in (**f**) shown as pink circles with error bars, indicating mean with standard deviation, data are mean ± standard deviation of six biological replicates. **h** High correlation between predicted output and cell death of the circuit. For box plots, the horizontal line represents the median value, the lower and upper quartiles represent the 25th and 75th percentiles, and the whiskers show the maximum and minimum values, white circles indicate the conductivities at 24 h post-induction in (**f**), Pearson $r = 0.92$. Source data are provided as a Source Data file.

coded genetic circuits for each state in plants[16,19] and paves the way for future work to establish a direct translation between the in vitro and in vivo data to enable direct prediction-to-phenotype engineering in plants. We also demonstrated that we can design circuits to control gene expression and cell death in response to different input chemical signals (Figs. 5 and 6). We envision that developing, designing and characterizing new sensors that can respond to different signals will further enhance the versatility and functionality of our circuits. While our current modeling framework is robust, it oversimplifies certain factors, such as cellular resource competition and protein turnover dynamics, that may influence circuit performance in plants. These aspects are particularly important in more complex circuits, where multi-layered gates require precise regulation and efficient signal propagation[31,32,35]. Future efforts should include incorporating protein degradation tags to mitigate accumulation issues[48], alongside multi-dimensional characterization of parts[49] and the integration of both mechanistic and machine learning models[50,51], to enhance model accuracy and optimize complex synthetic genetic circuits[52–56].

Our comprehensive framework integrates rapid validation through standardized characterization in protoplasts and transformation into stable plants for applications. This streamlined approach accelerates genetic circuit design and testing in plants to ~10 days, avoiding the extensive and lengthy iteration cycle of traditional plant transformation (>2 months per generation). Our rational and predictive framework can be applied to engineer plant traits for various purposes. For example, we can use our framework to design genetic circuits that can sense environmental signals (e.g., daily rhythms of agriculture, low oxygen in flooding) and regulate specific genes (e.g., floral regulator FLOWERING LOCUS T for flowering to maximize yield, *Arabidopsis* NAC domain-containing gene (ANAC102) for seed germination under low-oxygen stress) to modulate plant development or morphology, enhance plant growth, stress tolerance, or disease resistance[57,58]. We anticipate that it will contribute significantly to crop improvement, biotechnology applications, and fundamental studies of plant biology, facilitating advancements in precision agriculture and sustainable bioproduction.

Methods

Plasmids construction

To construct plasmids encoding genetic parts and genetic circuits, we used the plasmid backbone from a commercial vector pUC19. For the plasmids encoding genetic parts, the inserts were amplified using Phanta polymerase (Vazyme, P525) by custom-designed oligos and then ligated into the plasmid backbone using Golden Gate assembly with BsaI (NEB, R3733L) or BsmbI (NEB, R0739L) and T4 DNA ligase (NEB, M0202L). For the plasmids encoding genetic circuits, the genetic parts were amplified using Phanta polymerase and then ligated into the plasmid backbone using Gibson assembly (ClonExpress One Step Cloning Kit, Vazyme, C115). To construct the binary vector for transformation using *Agrobacterium tumefaciens*, we used the plasmid backbone from pCambia2300, and the genetic parts were amplified using Phanta polymerase (Vazyme, P525) and then ligated into the plasmid backbone using Gibson assembly (Vazyme, C115). All resulting products were transformed into *Escherichia coli* DH5α strains (AlpalifeBio, KTSM101L) and subsequently verified by Sanger sequencing. All plasmids associated with part IDs in the Inventory of Composable Elements that were used in this study are provided in Supplementary Data 1.

Modular cloning

The streamlined process of modular cloning is outlined in Supplementary Fig. 8, with detailed sequence information available in Supplementary Data 2–7. The vector backbone for rapid tests in *Arabidopsis* protoplasts used in this study comprises the high-copy replication origin from the commercial vector pUC19, the streptomycin resistance gene locus (SmR) in pL1 or kanamycin resistance gene locus (KanR) in pL2, and the LacZ gene locus from the pUC19 vector for blue-white screening.

In this study, Golden Gate[59] and Gibson assembly[60] were employed to develop a modular element assembly strategy. DNA elements were categorized into Level 1 (basic genetic elements stored in pL1) and Level 2 (assembled transcriptional units stored in pL2). Promoters and Ω-UTR-CDS-terminator components were seamlessly linked using defined prefixes/suffixes introduced through PCR primers and Type IIS enzymes (BsmBI, BsaI). Promoter variants labeled with '.1' and '.2' correspond to different replacements of the operator sequence within a 25-nt region between the TATA box and the transcription start site, with '.1' placed closer to the TATA box and '.2' closer to the TSS.

The prefixes and suffixes were introduced through PCR primers and Type IIS restriction enzymes (BsmBI, BsaI) for cloning. BsmBI Golden Gate cloning assembled elements into the pL1 vector, disrupting the LacZ gene for blue-white screening. Successful clones retained BsaI recognition sites upstream and downstream of the elements, enabling their transfer to the pL2 vector. pL1 and pL2 vectors were differentiated by resistance markers to prevent interference during clone selection. The TUs were stored on pL2 and equipped with Scar sequences at their ends for homologous recombination. Finally, TUs were assembled onto the final pCircuit vector using the Gibson assembly method, streamlining the construction of genetic circuits.

Plant materials and growth conditions

Protoplast experiments were conducted using the *Arabidopsis* Col-0. *Arabidopsis* plants for protoplasting and transformation through floral dip were cultivated in soil under a 16-h light/8-h dark cycle (22 °C/18 °C). For confocal analysis, plants were cultivated on half Murashige and Skoog (1/2 MS) medium (PhytoTechnology Laboratories, M524) agar plates, oriented vertically in a growth chamber with a 16-hour light/8-hour dark cycle (22 °C/18 °C).

Plant transformation

Agrobacterium tumefaciens GV3101 strain carrying pCambia2300 constructs was used for floral dip transformation to generate stable

Arabidopsis lines. Each construct contained a kanamycin-resistance gene for selection. T1 seeds were screened on kanamycin-containing 1/2 MS medium agar plates and subsequently propagated through T2 and T3 generations. Kanamycin-positive T2 seedlings were selected for treatment to test for induction.

Protoplasting and transfection

Protoplasts were isolated from leaves with a modified Tape-*Arabidopsis* Sandwich protocol[61], in which $1–2 \times 10^4$ protoplasts in 200 μL were added to a 2 mL round-bottom centrifuge tube with 30 μg of plasmid DNA (in 20 μL) before being mixed with 40% polyethylene glycol 4000 (PEG 4000, Sigma, 95904) and incubated for 10 min. After the reaction, 880 μL of W5 solution was slowly added, mixed, and centrifuged at 100 × *g* for 1 min using a horizontal rotor in a benchtop centrifuge. The supernatant containing the transformation solution was discarded, and the pellet was resuspended in an appropriate amount of WI solution. The resuspended solution was then transferred to a 48-well plate and incubated for 12–24 h to express the exogenous genes based on the experimental requirements. For experimental groups requiring induction of expression with naphthalene acetic acid (NAA, Macklin, N799072) or trans-Zeatin (tZ, Shanghai yuanye Bio-Technology Co., Ltd, S18004), after completing the above transfection process, induction was conducted within the 48-well plate. The concentrations of inducers, unless otherwise indicated in the figure legend, are provided in Supplementary Table 4. In the predictive design, the inputs of the sensors we used were measured in the same protoplast batch with the circuits. Different replicates were performed, and the number of replicates varied based on the specific experiment as indicated in the figure legends.

Calculation of RPU

After incubation, protoplasts were collected and lysed using 100 μL of lysis buffer (Yeasen, 11401ES60), followed by incubation on ice for 5 min to ensure complete lysis of the protoplasts. One-half of the lysates (45 μL) were transferred to a 96-well plate for LUC luminescence quantification. Another half of the lysates (45 μL) was transferred to another 96-well plate for GUS fluorescence quantification. The luciferase assay reagent was from Yeasen, while the GUS substrate was a homemade solution containing 4-methylumbelliferyl β-D-glucuronide (4-MUG, Yuanye, S11088)[62]. LUC luminescence was measured instantly after adding the luciferase assay reagent, and GUS fluorescence was measured after 1 hour of incubation at 37 °C, followed by stopping with Na_2CO_3 (Sigma, V900801). LUC and GUS were both quantified using a luminometer (BioTek Synergy H1), for GUS fluorescence the excitation light was set at a wavelength of 365 nm, and emission light was measured at 455 nm.

LUC/GUS values in a.u. were calculated as:

$$LUC/GUS = (LUC_{\text{test}} - LUC_0)/(GUS_{\text{test}} - GUS_0)(\text{a.u.}) \quad (1)$$

where LUC_0 and GUS_0 were the background luminescence and fluorescence values, respectively.

Then RPUs were calculated as:

$$RPU = (LUC/GUS)_{\text{test}}/(LUC/GUS)_{\text{P35S-200bp}} \quad (2)$$

Fold repression and fold induction calculation

The fold repression of repressors was calculated using the following equation:

$$\text{Fold change} = \text{Mean}\left(\text{RPU}_{\text{unrepressed}}\right)/\text{Mean}\left(\text{RPU}_{\text{repressed}}\right) \quad (3)$$

The mean values were derived from $n \geq 3$ experiments for both groups.

The fold induction of sensors was calculated using the following equation:

$$\text{Fold change} = \text{Mean}\left(\text{RPU}_{\text{induced}}\right)/\text{Mean}\left(\text{RPU}_{\text{uninduced}}\right) \quad (4)$$

The mean values were derived from $n \geq 3$ experiments for both conditions.

Response function modeling

Response functions of gates or sensors were fitted by Hill equation as:

$$f(x) = y_{\min} + (y_{\max} - y_{\min})\frac{K^n}{K^n + x^n} \quad (5)$$

where $f(x)$ is the output for gate/sensor, $y_{\min}$ is the minimum of the output, $y_{\max}$ is the maximum of the output, n is the Hill coefficient and K is the threshold level of input where the output is half-maximal. All fitting parameters are shown in Supplementary Tables 2 and 3.

Quantification and statistical analysis

Coefficient of variation (CV) was calculated as:

$$CV = \sigma/\mu \quad (6)$$

Here, σ is the standard deviation of our data, μ is the mean value of our data.

Quantitative data are shown in column-scattered plots or box plots (Prism). In column-scattered plots, the center, top, and bottom lines show the median, upper, and lower quartiles. In box plots, box plots indicate the median with lower and upper hinges corresponding to the 25th and 75th percentiles, respectively. The upper whisker extends from the upper hinge to the largest value, within a range of 1.5 times the interquartile range (IQR) from the hinge. Similarly, the lower whisker extends from the lower hinge to the smallest value, covering at most 1.5 times the IQR from the hinge.

Statistical analysis was performed with Student's *t*-tests (two-tailed, unpaired, or paired). Significance values are indicated on data plots. For multiple comparisons, one-way or two-way ANOVA was used and followed by a multiple comparisons test. The statistical details, including sample size for each measurement, were provided in figure legends and Source Data file. The significant differences were indicated as follows: ****, $p < 0.0001$; ***, $p < 0.001$; **, $p < 0.01$; *, $p < 0.05$; ns, not significant, $p > 0.05$.

Confocal microscopy

Plants after kanamycin selection were grown on plates for 10 days. For induction, seedlings were removed and immersed in a 1/2 MS liquid medium containing the inducers overnight. In this study, the final concentrations of NAA (5 mM NAA stock solution dissolved in ethanol) and tZ (5 mM tZ stock solution dissolved in DMSO) used for treating *Arabidopsis* seedlings were both 1 μM. Mock treatments were carried out in 1/2 MS liquid medium. Before imaging, roots were stained with 10 μg/mL propidium iodide (PI, Yeasen, 40710ES03) for 3 min to visualize cell outlines. Confocal images were captured by the Leica SP8 microscope in the GFP, RFP channels and the bright field.

Images of the quiescent center (QC) zone were taken using Leica SP8 with a 40× oil immersion objective, and the pinhole was kept at 1 airy unit. Images were processed using ImageJ. Identical imaging conditions were used for comparative samples. Where images are shown with brightness and/or contrast adjustments, comparative images were modified identically. For quantification, we drew a horizontal line at the upper boundary of the QC zone and included all cells from this line down to the tip, primarily encompassing the QC, columella, and lateral root cap cells. Relative GFP intensity was normalized to the highest ON state in each transgenic line.

Agroinfiltration and electrolyte leakage assay

Figure 6a outlines the procedure for agroinfiltration experiments conducted in *N. benthamiana*, generally following established protocols with minor modifications[63]. For the inducers treatments, we applied the infiltration method for gene expression induction, which has been used in several studies[64–66]. Specifically, agrobacteria containing the circuit plasmid were resuspended in 10 ml of infiltration buffer, which contained 10 mM $MgCl_2$ (Sigma, V900020), 10 mM 4-morpholineethanesulfonic acid (MES, Sigma, V900336) and 0.2 mM acetosyringone (Yeasen, 60326ES03). A volume of 1 mL of bacteria with an OD600 = 0.2 was taken from the same bacterial culture and syringe-infiltrated into the adaxial sides of *N. benthamiana* leaves at four distinct, equal-sized areas. After 24 h post-infiltration, the infiltrated leaves were treated with different hormonal inputs to test the OR logic operation. Specifically, we infiltrated water, NAA, tZ, and both hormones into separate, spatially distinct regions on the same leaf, with sufficient spacing between each treatment to minimize the risk of hormone diffusion between regions. Both NAA and tZ were maintained at final concentrations of 1 µM. Observations and imaging under white light or long-wave ultraviolet were conducted 2 days post-infiltration. Three experimental replicates were performed for each experiment.

For electrolyte leakage assays, independent *N. benthamiana* leaves were injected in four distinct areas with the same volume of *Agrobacterium* solution from the same culture. After 24 h post-infiltration, the infiltrated leaves were induced with different inputs, as depicted in the figures. Three discs from each area were collected and floated on 50 mL ddH_2O for 30 min and then were transferred into tubes with 5 mL ddH_2O. Conductivity was measured using an Orion Conductivity Meter (Thermo Scientific) at the indicated time points after induction.

Data visualization and figure preparation

GraphPad Prism 7.00 was used for both statistical analysis and data visualization. For tasks such as fitting transfer functions, plotting response curves, illustrating gene circuit predictions, and generating heatmaps, Python scripts utilizing the matplotlib library were employed for modeling and visualization. Microscope image processing was conducted using LAS X (Leica Microsystems) and ImageJ. Figures were created in Adobe Illustrator CC 2019.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All plasmids with part IDs cataloged in the Inventory of Composable Elements (ICE, available at https://registry.jgi.doe.gov/) are detailed in Supplementary Data 1. Source data are provided with this paper.

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Author contributions

S.Z. conceptualized and supervised the project. C.K. and S.Z. participated in all experiments, design, and data analysis, Y.Y. and T.Q. contributed to tobacco HR phenotype studies. C.K. and S.Z. wrote the paper with help from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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